Quinn Tyler Jackson

CSci FIScT SMIEEE SMACM FRSA

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Before my current role as Chief Scientist – Mathematics, I was a Chief Scientist and entrepreneur for a decade, working in the areas of bioinformatics, SEC compliance, automated medical and financial report analysis, and NLP; before that, I was Chief Architect at Faronics Corporation. My work in adaptive parsing and grammar theory and practice has been cited in papers, dissertations, and various other scholarly works. I am a Chartered Scientist (UK), a Fellow of the Institute of Science and Technology (UK), a Senior Member of both the IEEE and ACM, a Member of the London Mathematical Society, the American Mathematical Society, and a Fellow of the Royal Society of Arts.

WORK HISTORY

2020-present Chief Scientist – Mathematics, Pattern Computer Inc.

Reporting directly to the CEO in an innovative industrial setting, I am responsible for the oversight of all mathematics (applied and theoretical) at Pattern Computer and serve to integrate the scientific, mathematical, and engineering aspects of the engineering department under the roof of applied mathematics and machine learning.

- Conceived, designed, implemented, optimized, and researched new proprietary machine learning algorithms, applied mathematics, and model building system and subsystems, from concept to measured results, increasing our tangible industry-competitive technological assets, resulting in 20+ detailed technical reports outlining technological improvements and results in the areas of symbolic regression, new ML algorithms and enhancements, and explainable AI
- Developed and promoted scientifically sound technology-specific methodologies across suite of proprietary tools and technologies.
- Participated in, gave input on, and reviewed interdisciplinary mathematics and bioinformatics team discussions

2018-2020 Senior Bioinformatics Technologist, Pattern Computer Inc.

- Co-invented/developed a *de novo* drug combination for Triple-Negative Breast Cancer discovered via high-dimensional pattern discovery. (See Publications – Patents)
- Conceived, designed, prototyped, and validated over 30 trade secret and/or patent-pending machine learning technologies and algorithmic enhancements
- Validated, optimized, instrumented, and productized key proprietary algorithms
- Participated in, gave input on, and reviewed interdisciplinary mathematics and bioinformatics team discussions and summits

2016-2018 Senior Software Engineer II, Synopsys Canada Inc.

- Led productization of proof-of-concept security analysis subsystem
- Researched and reported on feasibility of introducing new functionality to end-user workflow subsystem
- Designed, implemented, and tested various MISRA C++ 2008 and MISRA C 2012 rule compliance checkers
- Implemented new quality analysis checker and supporting subsystem
- Reviewed and tested various team changes and enhancements to the front and back ends of the static analysis engine in the C/C++, Java, C#, and Scala static analysis subsystems

2009-2016 Chief Scientist, Q Agamemnon Technologies

- Researched, designed, developed, optimized, and innovated in NLP space (document segmentation, discourse analysis, POS tagging, entity recognition, Ontology Population, syntactic and semantic disambiguation, relationship discovery, analysis, and resolution)
- Researched, designed, developed, and innovated in bioinformatics and terminology discovery in de-personalized medical records, chemical (pharmaceutical) patents, medical interaction relationship discovery, analysis, and normalization
- Designed, implemented, validated, and optimized systems (C++, SQL, and proprietary technologies)
- Collaborated directly with clients and partners on R&D of various grammars in the areas of bioinformatics in areas of genetic research, genomic secondary structures, cancer pathology reporting, USDA classification, pharmaceutical patent analysis, SEC, and other international trade compliance issues

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2005-2009 Chief Scientist, Thothic Technology Partners, LLC

- Planned and implemented R&D of new technologies in the areas of chemical name analysis, bioinformatics and genomic secondary structure recognition, clinical terminology recognition in de-personalized medical records, molecular structure normalization
- Strategized with partners on technology and business and research direction and presentation
- Prepared and presented many demonstrations of key in-house technologies to government and private industry in health, mathematics, and other industries and interests
- Assisted industry and government clients and researchers with system integration and testing/benchmarking
- Collaborated directly with clients on R&D of various grammars in the areas of genetic research, SEC and other international trade compliance issues

2003-2005 Chief Architect, Faronics Corporation

- Negotiated system requirements with various stakeholders for company's proposed multi-million-dollar enterprise system
- Proposed, built, and organized core team of developers specifically tasked to implement core architecture in C++ who reported to me
- Evaluated core platform technologies (COM, WMI, TCP/IP, MFC/ATL, et al.) and core company asset technologies (Deep Freeze) for suitability to purpose
- Mentored team and non-team members in various coding, design, and implementation techniques and strategies (design patterns, C++ techniques, OOD and implementation)

2001-2003 Software/Parsing Consultant, Jackson Solutions

- Proposed, designed syllabi and exams, taught, and examined undergraduate level courses in software design, software documentation, network security, and electronic-age legal issues
- Designed, implemented, and tested back-end code framework for proprietary client computer language prototype
- Designed, implemented, and tested LALR(k) grammars and back-end reduction code for C#, Perl, HTML, and various proprietary little languages
- Researched and implemented adaptive grammar and parsing algorithms

1997-2001 Research Programmer, Hyper Technologies Inc.

C++ coding, system design, research, and exploratory programming.

- Designed, implemented, and brought through various released versions a lock down system that was installed on thousands of public access systems
- Assisted with design/implementation of various algorithms for system lock down and restoration

1996 Software Developer, Lambda Systems Ltd.

System design, research, exploratory programming, and consulting.

- Researched, designed, implemented, tested, and delivered public access computer system (PACS) kiosk control and lockdown technology (iKIOSK) that was directly and profitably taken to market by client
- Designed, consulted on, implemented, and tested visual patient record UI component in major client medical care tracking system

FORMAL QUALIFICATIONS

- Chartered Scientist (via Institute of Science & Technology, UK) – 1 February 2018

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PROFESSIONAL AND LEARNED ASSOCIATIONS

- Fellow, Institute of Science & Technology (UK)
- Fellow, Royal Society for the Encouragement of Arts, Manufactures and Commerce
- Senior Member, Institute of Electrical and Electronics Engineers (IEEE)
- Senior Member, Association for Computing Machinery (ACM)
- Member, American Mathematical Society (AMS)
- Member, London Mathematical Society (Reciprocity via AMS)
- Member, Writers' Union of Canada
- Member, IEEE Computational Intelligence Society

PUBLICATIONS (COMPUTING/MATHEMATICS/LINGUISTICS ONLY)

Patents (Granted and Pending)

- "System and Methods for Generating a Reduced-Order Surrogate Model using Genetic Algorithms," inventors: **Quinn Jackson**, Matt Keener, Gavin Dean, Filed 17 March 2021 (PENDING).
- "Therapeutic Compositions and Methods for Treating Cancers," inventors: Nidhi Singh, Meenakshi Venkatasubramanian, **Quinn Jackson**, PCT/US2020/038684, Filed 19 June 2020
  (GRANTED: US **12,521,394**; EU: **EP3999065**; CHINA: **ZL202080045051.4**; INDIA: Granted **07Aug2025**)

Books

- *Adapting to Babel: Adaptivity & Context-Sensitivity in Parsing (from $a^n b^n c^n$ to RNA),* Ibis Publishing, Plymouth, Massachusetts, Mar. 2006.
- *Parsing with Visual Parse++,* SandStone Technology Inc., 2001.

Papers

- "Explainable molecular stratification and co-regulation pattern identification in hepatocellular carcinoma single cell transcriptomics data," w/ Angelica M. Walker, Nidhi Singh, Luca Pion-Tonachini1. James B. Brown, **Quinn Jackson**, Ken Kreutz-Delgado, Ty Carlson, Alejandro Wolf-Yadlin, *in submission*, 2024.
- "Machine learning prediction of adenovirus D8 conjunctivitis complications from viral whole genome sequence," w/ Kenji Nakamichi, Lakshmi Akileswaran, Thomas Meirick, Michele D. Lee, James Chodosh, Jaya Rajaiya, David Stroman, Alejandro Wolf-Yadlin, **Quinn Jackson**, W. Bradley Holtz, Aaron Y. Lee, Cecilia S. Lee, Russell N. Van Gelder, *Ophthalmology Science*, 8 May 2022, available at URL: https://doi.org/10.1016/j.xops.2022.100166
- "Pattern Discovery Beyond Published Results: Three Papers," Pattern Computer Inc., June 2021.
- "Learning from learning machines: a new generation of AI technology to meet the needs of science," preprint, co-authored with 35 others, 27 November 2021, available at URL: https://doi.org/10.48550/arXiv.2111.13786
- "On Einstein's Razor: Telesis-Driven Introduction of Complexity into Apparently Sufficiently Non-Complex Linguistic Systems," *Progress in Complexity, Intelligence, and Design*, Vol. 4.2 Nov. 2005
- "Some Theoretical and Practical Results in Context-Sensitive and Adaptive Parsing," *Progress in Complexity, Information, and Design*, Vol. 1 No. 4, Dec. 2002.
- "Efficient Formalism-Only Parsing of XML/HTML Using the §-Calculus," *Noesis-E*, Vol. 2 No. 2, June 2002. (Reprinted in *ACM SIGPLAN Notices*, Feb. 2003)
- "An Introduction to Adaptive Grammar Theory and Practice," *Noesis-E*, Summer, 2002.
- "Adaptive Predicates in Empty-Start Natural Language Parsing," with Christopher Michael Langan, *Noesis-E*, Vol. 1 No. 6, Nov. 2001.
- "An Introduction to PAISLEI," *Noesis-E*, Mar. 2001.
- "Adaptive Predicates in Natural Language Parsing," *Perfection,* Vol. 1 No. 4, Apr. 2000.
- "Disambiguation as a Quantifiable Computational Process," *Perfection*, Vol. 1 No. 3, Mar. 2000.

Articles

- "Reflective Practice: On Diversity & Inclusion in Science & Technology," *The Tech Magazine of the IST*, Vol. 2 No. 1, March 2021.
- "Jackson's #ScientistJourney to Software Engineer," *IST Newsletter*, November 2018.
- "Linguistic Time-Space," *Perfection*, Vol. 1 No. 8, Sept. 2000.
- "Deciphering the Consequences of Babel," *Ubiquity*, Vol. 1 No. 4, Apr. 2000.

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EXPERTISE KEYWORDS MAPPED BY ACM DOMAINS

{ACM:Software and its engineering~Software organization and properties: design patterns, software development, business requirements analysis, systems analysis, system architecture, documentation};

{ACM:Software and its engineering~Programming languages: C++, SQL, JavaScript, R, Python};

{ACM:Software and its engineering~Operating systems: Windows, Linux};

{ACM:Theory of computation~Design and analysis of algorithms: algorithm research and implementation, algorithm optimization, evolutionary algorithms, submodular optimization, peephole optimization};

{ACM:Mathematics of computing~Mathematical foundations: set theory, category theory, lattice theory, sheaf theory, universal algebra, abstract algebra, propositional calculus, formal proof theory, combinatorics};

{ACM:Mathematics of computing~Continuous mathematics: measure theory, Lebesgue integration};

{ACM:Mathematics of computing~Probability and statistics: probability theory, categorical data analysis};

{ACM:Mathematics of computing~Graph theory: graph theory, probabilistic graph modeling, social network analysis};

{ACM:Mathematics of computing~Information theory: information theory};

{ACM:Computing methodologies~Machine learning: symbolic regression, counterfactuals and causal inference};

{ACM:Computing methodologies~Natural language processing: natural language processing, parsing, computational linguistics, corpus linguistics};

{ACM:Theory of computation~Formal languages and automata theory: grammar theory, adaptive grammar theory, automata theory, language theory, X-bar theory, theta role theory};

{ACM:Applied computing~Life and medical sciences~Bioinformatics: bioinformatics};

{ACM:Software and its engineering~Software notations and tools: static analysis, compiler construction};

{ACM:General and reference~Cross-computing tools and techniques: scientific research and experimental development}

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